

Clara Marcille, PhD in Computer Science

✉ clara.marcille@gmail.com

🌐 <https://clara-marcille.github.io/perso/>



Formation

- 2026-  Postdoc at LIMOS, Clermont-Ferrant, France.
- 2025 – 2026  **Temporary Assistant for Teaching and Research** ENS de Lyon.
- 2022 – 2025  **PhD Student** LaBRI, Université de Bordeaux. PhD supervised by Julien Bensmail and Hervé Hocquard.
- 2021 – 2022  **Preparation for the Agrégation d'Informatique** Computer Science department of ENS de Lyon.
- 2020 – 2021  **Master 2 : Computer Science** at ENS de Lyon, France. **Stage** : Going wide with the 1-2-3 Conjecture. **Supervisors** : Julien Bensmail et Hervé Hocquard.
- 2019 – 2020  **Master 1 : Computer Science** at ENS de Lyon, France.
- 2018 – 2019  **License : Computer Science** at ENS de Lyon, France.

Skills

- Academic  Graph Theory, Coding, Combinatorics, Parameterized Complexity.
- Langues  French (mother tongue), English, German.
- Code  C, C++, Python, SQL, $\text{\LaTeX}$, OCaml, HTML, CSS

Talks

- 2022, 2023  **The Weak (2,2)-Labelling Problem for graphs with forbidden induced structures.**
JGA 2022, ENS Ulm, France.
Caldam 2023, IIT Gandhinagar, India.
- 2023, 2024  **On inducing degenerate sums through 2-labellings.**
Seminar GT-GO, LaBRI, France.
Mathematics Weekly Seminar, IIT Dharwad, India.
- 2023  **Adding direction constraints to the 1-2-3 Conjecture.**
Poster session for the annual meeting of GdR-IFM, IRIF, France.
- 2023, 2024  **Monitoring edge-geodetics of graphs.**
Seminar GT-GO, LaBRI, France.
Seminar Alcoloco, LIMOS, France.
- 2023  **Monitoring edge-geodetic sets on oriented graphs**
JGA 2023, Lyon 1, France.
- 2024  **From Antimagic to Equitable Labellings**
Poster session for the days of EDMI, Université de Bordeaux, France.
CombALGO team day, LaBRI, France.
IWOCA 2024, Ischia, Italie.
-  **Strongly locally irregular graphs and decompositions**
JGA 2024, Université de Bourgogne, France.

Talks (next)

- 2025  **Graph Irregularity via Edge Deletions**
Graph Seminar of IRIF, IRIF, France. Seminar GT-GO, LaBRI, France.
Distinction and Detection Problems in Graphs
PhD Defense, LaBRI, France.
- 2026  **Structural Parameterizations of Path Cover**
Graph Seminar of CRISTAL, Lille, France.

Responsabilités collectives et activités de diffusion

- 2022  **Journées Math à Modeler:** Designing combinatorial games to introduce combinatorics to high-school-level students, Grenoble.
- 2023  **Mathematical Games Salon:** Animating a booth on mathematical games and combinatorics for the Math à Modeler project for the large public, Paris.
- 2023-2024  **Organizing the seminar for interns:** finding speakers, organizing and chairing the sessions for the bi-weekly seminar of LaBRI between May and July, to introduce the interns to the various fields of research in the lab.
-  **Secretary then President of the PhD students association:** Administrative responsibilities, and organization of social events for the non-permanent members of LaBRI.
- 2023-2025  **Council of the Science, Engineering and Numerics departement of University of Bordeaux:** PhD students representant.
-  **Organizing the GT-GO seminar:** finding speakers and chairing the weekly seminar of the graph theory team in LaBRI.
-  **Member of the workshop for mental health at work:** Member of the lab trained in preventing mental health problems in the workplace. Organizing group discussion and support to improve the work environment in LaBRI.
- 2024-2025  **DEI mission in LaBRI:** organizing awareness events in LaBRI, as well as chosen mixity events to exchange between minorities in the lab.
- 2024-  **DEI comitee of GT graphes for GdR-IFM:** comitee for diversity, equity and inclusion initiatives at national level within the graph community in France.

Other experiences

Prizes and Awards

- 2023  **Best Student Presentation Award**, Caldam 2023, Gandhinagar, India.
- 2024  **Raman-Charpak Fellowship**, CEFIPRA.
-  **Best Student Paper Award**, Caldam 2024, Bilhai, India.

Certifications

- 2016  **Certified B2 in German** from the Goethe Institute.
- 2021  **Certified C1 in English** from CAE.

Research visits

- 2023  International visit (one month), IIT Dharwad, India.
-  GDR-IFM exchange program for PhD. students (one week), LIMOS, Clermont-Ferrand, France.
- 2024  International visit (two months), IIT Dharwad, India.
-  attended EPIT 2024 summer school, Aussois, France.

Other experiences (next)

Reviews

- 2023 – 2025  Discrete Applied Mathematics.
- 2023  Caldam 2024.
- 2024  Ars Combinatoria.
-  Acta Informatica.
-  Caldam 2025.

Other trainings

- 2023  Mental health awareness (20h).
-  Management (12h).
- 2025  Data Vizualisation (12h).

Publications

International journals

- 1 J. Bensmail, T. Filasto, H. Hocquard, and C. Marcille, “Irregularity notions for digraphs,” *Graphs and Combinatorics*, vol. 41, no. 3, p. 60, 2025.
- 2 J. Bensmail, H. Hocquard, and C. Marcille, “Adding direction constraints to the 1-2-3 conjecture,” *Theoretical Computer Science*, vol. 1025, p. 114 985, 2025.
- 3 J. Bensmail, H. Hocquard, C. Marcille, and S. Meyer, “On 1-2-3 conjecture-like problems in 2-edge-coloured graphs,” *Discrete Mathematics*, vol. 348, no. 4, p. 114 368, 2025.
- 4 J. Bensmail, C. Marcille, and M. Orenga, “Pushing vertices to make graphs irregular,” *Discrete Mathematics & Theoretical Computer Science*, vol. vol. 27:3, 2025.
- 5 T. Das, F. Foucaud, C. Marcille, P. Pavan, and S. Sen, “Monitoring arc-geodetic sets of oriented graphs,” *Theoretical Computer Science*, vol. 1031, p. 115 079, 2025.
- 6 F. Foucaud, C. Marcille, Z. M. Myint, R. Sandeep, S. Sen, and S. Taruni, “Bounds and extremal graphs for monitoring edge-geodetic sets in graphs,” *Discrete Applied Mathematics*, vol. 366, pp. 106–119, 2025.
- 7 J. Bensmail, H. Hocquard, and C. Marcille, “On inducing degenerate sums through 2-labellings,” *Graphs and Combinatorics*, vol. 40, no. 2, p. 23, 2024.
- 8 J. Bensmail, H. Hocquard, and C. Marcille, “Going wide with the 1-2-3 conjecture,” *Discrete Applied Mathematics*, vol. 320, pp. 15–32, 2022.

Conferences

- 1 J. Bensmail, N. Catherinot, F. Fioravantes, C. Marcille, and N. Oijid, “Graph irregularity via edge deletions,” in *WALCOM: Algorithms and Computation*, 2026, pp. 353–367.
- 2 C. Marcille and N. Oijid, “An algorithm for monitoring edge-geodetic sets in chordal graphs,” in *to appear in IWOCA*, 2026.  URL: <https://arxiv.org/abs/2602.03288>
- 3 J. Bensmail and C. Marcille, “An improved bound for equitable proper labellings,” in *Proc. of IWOCA’24*, 2024, pp. 82–97.
- 4 F. Foucaud, C. Marcille, Z. M. Myint, R. B. Sandeep, S. Sen, and S. Taruni, “Monitoring edge-geodetic sets in graphs: Extremal graphs, bounds, complexity,” in *Proc. of CALDAM’24*, 2024, pp. 29–43.
- 5 J. Bensmail, H. Hocquard, and C. Marcille, “The weak (2, 2)-labelling problem for graphs with forbidden induced structures,” in *Proc. of CALDAM’23*, 2023, pp. 204–215.